

Program	Bachelor of Technology (B.Tech.)	Semester - 5
Type of Course	Core Courses	
Prerequisite	03070403-T - INTRODUCTION TO INTERNET AND HTML	
Rationale	The purpose of this course is to be able to explain, configure, verify, and troubleshoot complex computer networks at an introductory level.	
Effective From A.Y.	2024-25	

Teaching Scheme (Contact Hours)				Examination Scheme				
Lecture	Tutorial	Lab	Credit	Theory Marks		Practical Marks		Total Marks
				T	T	P	P	
3	0	2	4	70	30	-	50	150

SEE - Semester End Examination, T - Internal Theory, P - Internal Practical

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	Introduction to Computer Network Network History, Introduction of Network, Uses of computer network, Application of Network, Network hardware, Network software, Network Simulator, Network, Differentiate various Line Configuration, use of Server, use of various types of server and their role, Types of Network, Internet, Intranet, Extranet, Protocols, Connection Oriented and Connection less service, Peer-Peer Network, Client -Server Network	10	15
2	The Reference Model OSI Model, TCP/IP Model, Comparison of TCP/IP and OSI, Topology, types of topology, Design and their implementation, Types of switching network, Circuit switching, Packet switching, Message switching, MAC address, Mobile telephone system, Line Configuration, Multiplexing, Simple Duplex, Half Duplex, Full Duplex, Protocols- stop & Wait, Sliding Window	15	20
3	Transmission Medias and IP address Introduction of Transmission Medias- Twisted, Coaxial, Fiber optical cables, Radio waves, Infrared, Microwave, Laser, Use and application of medias, connectors, Ethernet cabling, Comparison of wired and wireless media, IP protocol, Subnet masking, Types of IP and classification, IP address scheme, IP V4, IP V6, IPV4 header, Routing and Routing Protocols,	15	20
4	Network Devices and Flow Control Network Devices Network devices and their types: Repeaters, Hubs, Switches, Router, Bridges, Brouter, Gateways, Comparison and use, LAN implementation and requirement, Flow Control, Flow Control techniques, Data Error and types, Error correction and detection techniques, Hamming code, Hamming distance	10	20
5	The Application Layer DNS, DNS name space, Resource records, Name servers, URL, Electronic mail, Mail protocols, Email architecture and services, User agent, Message formats, Message transfer, Final delivery, World Wide Web Architectural overview, File Transfer Protocol and Trivial File Transfer Protocol, HTTP, HTTPS, Network Security, Firewall, Types of Firewall, TCP, UDP, Network commands	10	25
Total		60	100

Suggested Distribution Of Theory Marks Using Bloom's Taxonomy					
Level	Remembrance	Understanding	Application	Evaluate	Create
Weightage	20	15	20	20	25

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:

CO1	Describe the components and infrastructure that form the basis for most computer networks
CO2	Describe the technical aspects of Computer Network on the Internet.
CO3	Understand the use of various Network components and Transmission Media.
CO4	Understand routing strategies for an IP based networking infrastructure.
CO5	Configure Different topologies in Cisco Packet tracer

CO PO Mapping

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5
PO - 1	3	3	3	3	2
PO - 2	2	2	2	3	3
PO - 3	1	2	2	2	3
PO - 4	1	2	2	3	2
PO - 5	1	2	2	2	3
PO - 6	2	2	1	1	1
PO - 7	1	1	1	1	1
PO - 8	1	2	1	1	1
PO - 9	1	2	2	2	2
PO - 10	2	2	1	1	2
PO - 11	1	1	1	1	2
PO - 12	2	2	2	2	3

Reference Books

1.	Computer Network By Andrew S. Tanenbaum Pearson
2.	Introduction to Data Communication and Networking (TextBook) By Behrouz Forouzan TMH



List of Practical

1.	Write down basic Networking commands.
2.	Configure Line (Bus) Topology in Cisco Packet Tracer.
3.	Configure Ring Topology in Cisco Packet Tracer.
4.	Configure Star Topology in Cisco Packet Tracer.
5.	Configure Mesh Topology in Cisco Packet Tracer.
6.	Configure Tree Topology in Cisco Packet Tracer.
7.	Configure Hybrid Topology in Cisco Packet Tracer.
8.	Write down steps to configure Local Area Network.
9.	Give Classification of Transmission Medias with diagram.
10.	Give IP address Classification.